

Computer Vision Engineer Take-Home Assessment

Retail Behavioral Analytics

Purpose: Design and implement an end-to-end video analytics pipeline that measures customer interest, shelf engagement, and staff-customer interaction from retail CCTV footage. The assessment emphasizes practical computer vision reasoning, temporal event logic, reproducibility, and clear technical communication.

Deadline: Within 5 calendar days of receiving the assignment. Submit by 23:59 (GMT+8) on the due date.

1. Assessment Overview

A retail store would like to use video analytics to better understand customer behavior and engagement. You will be provided with 2 CCTV video snippets capturing the storefront and interior respectively. Your task is to develop a solution that extracts three categories of behavioral analytics from these videos.

The objective is not necessarily to achieve perfect accuracy. We are interested in how you:

- Analyze an open-ended computer vision problem and turn ambiguous business requirements into measurable events.
- Select and integrate appropriate models, algorithms, and heuristics into a coherent pipeline.
- Build a reproducible implementation with clear configuration, documentation, and outputs.

You are free to choose the technical approach. The assessment intentionally defines the desired behavioral outcomes more precisely than the implementation method so that you can demonstrate engineering judgement.

2. Input Data and Scene Setup

You will be provided with 2 video snippets captured from a single fixed CCTV camera overlooking the storefront and interior of the retail store.

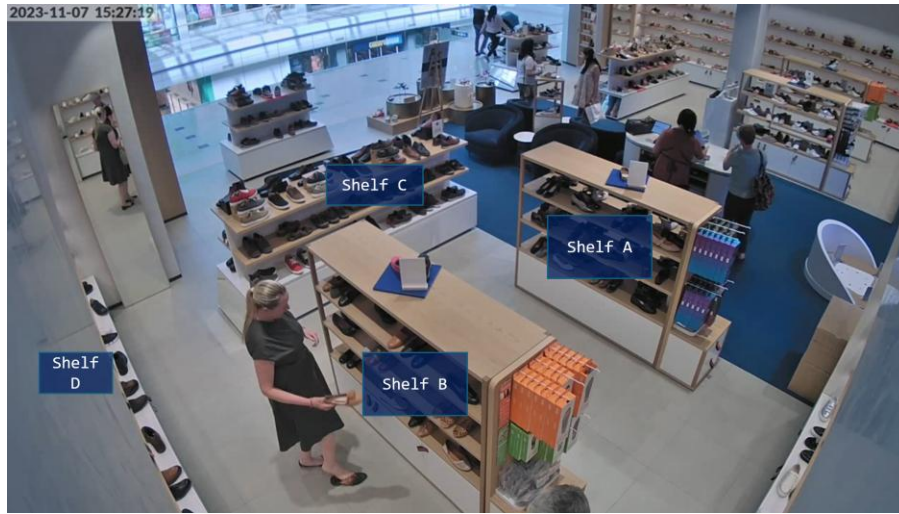


Figure 1. Example retail interior view. The 4 shelves/areas for which customer interest should be measured indicated as shelf A-D.

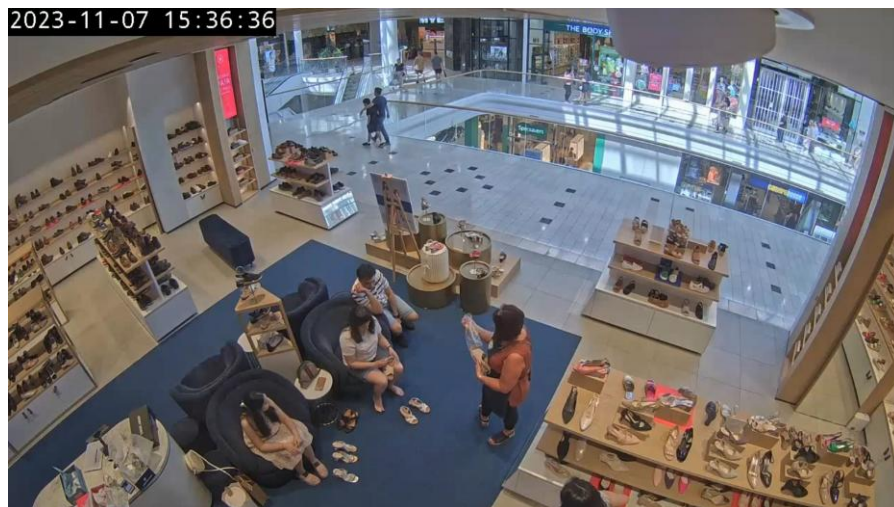


Figure 2. Example retail entrance view.

Each video snippet should be evaluated independently. Identity persistence across different video snippets is not required.

Input video specifications:

- Number of video snippets: 2
- Video format: .mp4
- Resolution: 1280 x 720
- File Names:
 - entrance.mp4
 - interior.mp4

3. Required Analytics

Any criteria, thresholds, or time-based rules introduced by the candidate should include a brief comment near the relevant code explaining the rationale behind the chosen values or logic. The `README.md` should clearly document each implementation in detail, including its purpose, methodology, decision rules, assumptions, parameters, and results.

All implemented outputs must also be clearly visualized in the final video output. Counts should be displayed at the top-left corner of the video, all machine-learning detections should be visibly annotated, and interaction-related results should be presented clearly so that the detected interactions can be easily understood.

Task 1 - Store Interest and Conversion

Measure how many passers-by show interest in the store and determine whether those interested people subsequently enter the store or continue walking past it.

Using `entrance.mp4`, report:

1. **Total Interested:** The number of unique people who showed interest in the store.
2. **Interested Entered:** The number of those interested people who subsequently entered the store.
3. **Interested Passed By:** The number of those interested people who passed by without entering.

The three reported counts must be clearly displayed as text overlays in the output video generated from `entrance.mp4`.

The same results must also be provided in a CSV file, with each metric clearly labelled and its corresponding count recorded.

Minimum definition of store interest

A person should be considered interested when there is visual evidence that they direct their attention toward the store. This should not be determined solely by whether the person stops in front of the store; other observable behaviors, such as looking toward the storefront, turning their head or body toward it, slowing down, or approaching the entrance, may also indicate interest. Candidates should define a reasonable set of criteria and clearly explain the reasoning behind their decisions.

Entered store

A person should be classified as having entered the store when they cross the entrance boundary and continue into the store. Candidates should be prepared to explain any thresholds or temporal rules used to make this determination.

Passed by

An interested person should be classified as passing by when they continue past the storefront without crossing the entrance boundary into the store.

Expected Output:

1. An annotated version of `entrance.mp4` with the three final counts clearly displayed as text overlays
2. CSV file reporting the labelled totals for Total Interested, Interested Entered, and Interested Passed By.

Task 2 - Per-Shelf Customer Interest

For each designated shelf in `interior.mp4`, determine the total number of customer interest events observed for that shelf.

Minimum definition of shelf interest

A customer is considered interested in a shelf when there is evidence that they direct their attention toward it for a sustained period. When a customer enters a region between two shelves, candidates should determine which shelf the customer is interacting with based on the available visual evidence.

Candidates should apply consistent criteria across the entire video and be prepared to explain the reasoning behind their decisions.

Repeated interest and double counting

A customer continuously interacting with the same shelf for several seconds should be counted as one interest event, rather than being counted repeatedly.

The solution should define how it distinguishes a continuous interest episode from a later, separate return to the same shelf. A customer returning to the same shelf after the previous interaction has ended should be counted as a new event.

Visualization requirements

The output video should clearly show:

- Which shelf each interested customer is interacting with
- The duration of the current shelf interaction
- A clear visual association between the customer and the corresponding shelf
- The cumulative number of interest events recorded for each shelf

Expected Output:

1. An annotated version of interior.mp4 that clearly identifies the shelf each customer is interacting with, displays the duration of each active interaction, and shows the cumulative interest-event count for every designated shelf
2. A CSV file containing the clearly labelled final count for each shelf.

Task 3 - Staff-Customer Interaction

Using `entrance.mp4`, calculate the average number of customer interaction sessions per staff member observed in that snippet.

Staff identification

Staff members can be visually distinguished from customers by the aprons they wear. The solution may use visual appearance, behavior, spatial context, or any other reasonable approach to distinguish staff members from customers.

For this assessment, a staff member should be treated as the same staff instance for as long as they remain within the camera view. If they leave the view and later re-enter, they may be treated as a new instance. Re-identification across separate appearances is optional.

All detected staff instances should be included when calculating the average number of interactions, including staff members with zero detected interactions.

What constitutes an interaction?

An interaction should involve evidence that a staff member and customer are actively engaging with one another. Candidates should define reasonable criteria and temporal rules for determining when an interaction begins and ends, and should clearly document and be prepared to explain the reasoning behind these decisions.

Interaction sessions and repeated interactions

The required metric is based on **interaction sessions**, rather than the number of unique customers interacted with.

A continuous period of interaction between a staff member and customer should count as one interaction session. If the interaction ends and the customer later returns to interact with the same staff member again, the later interaction should be counted as a separate session.

Similarly, interactions involving the same customer at different times may contribute multiple interaction sessions where they represent distinct periods of engagement.

Metric calculation example

Staff instance	Interaction sessions
Staff 1	5
Staff 2	4
Staff 3	0
Average	$(5 + 4 + 0) / 3 = 3.0$

Shared visualization requirement

The Task 3 results must be displayed in the same annotated output video generated from entrance.mp4 for Task 1. Candidates should not produce a separate video for Task 3.

The shared output video should clearly show:

- Detected staff instances
- Active staff-customer interaction sessions
- The cumulative interaction-session count for each staff member
- The final average number of customer interaction sessions per staff member
- The three Task 1 metrics: Total Interested, Interested Entered, and Interested Passed By

Expected Output

1. A single annotated version of entrance.mp4 containing the required visualizations and results for both Task 1 and Task 3.
2. A CSV file containing each staff instance, including staff members with zero interactions, the number of interaction sessions recorded for each staff member, and the final average number of customer interaction sessions per staff member.

4. Deliverables

Candidates must submit annotated output videos and CSV files containing the results for all three tasks. The visualizations should clearly demonstrate how the reported results were obtained and should remain readable throughout the relevant portions of each video.

4.1 Annotated Videos

Candidates must submit:

- One annotated version of entrance.mp4 containing the visualizations and results for both **Task 1 – Store Interest and Conversion** and **Task 3 – Staff-Customer Interaction**
- One annotated version of interior.mp4 containing the visualizations and results for **Task 2 – Per-Shelf Customer Interest**

The videos must clearly display all required detections, classifications, interaction durations, cumulative counts, and final results. All text overlays and visual associations should be clear and readable.

4.2 CSV Files

Candidates must submit CSV files containing the final numerical results for all three tasks:

Task 1: Total Interested, Interested Entered, and Interested Passed By

Task 2: Total customer interest events for each designated shelf

Task 3: Interaction-session count for each staff instance and the average number of customer interaction sessions per staff member

Staff instances with zero interactions must be included in the Task 3 results. All values in the CSV files must match the results shown in the annotated videos.

4.3 Documentation

Candidates must submit a clear and complete `README.md` containing:

- The methodology used for each task
- The criteria, thresholds, temporal rules, and formulas applied
- The technologies, models, libraries, and tools used
- The reasoning and justification behind key technical and design decisions
- Setup and execution instructions
- The expected output files and folder structure
- Any assumptions, limitations, and known failure cases

The `README.md` should be detailed enough for reviewers to understand how the solution works, reproduce the results, and assess the candidate's technical reasoning.

5. Technical Constraints and Allowed Tools

Area	Policy
Pretrained models	Allowed. Publicly available pretrained model weights may be downloaded during environment setup
Open-source libraries	Allowed, with clear installation and setup instructions provided in the README.
External / paid cloud APIs	Not allowed. All inference and processing of the provided video footage must execute locally using installed dependencies and/or locally available model files. The provided video footage must not be uploaded to external inference or processing services.
Coding Assistants during Development	Coding assistants are allowed. Candidates remain fully responsible for the submitted solution and must ensure that they understand the generated or assisted code, methodology, and technical decisions. Use of a coding assistant does not reduce the requirement for the solution to be reproducible, well documented, and independently explainable by the candidate.
Execution environment	Provide a Dockerfile and any supporting dependency or configuration files required to reproducibly build and run the submitted solution.

6. Submission Requirements

Submit a ZIP archive containing the source code, configuration files, documentation, Dockerfile, environment/dependency specifications, and generated outputs required to reproduce your results.

Publicly available pretrained model weights do not need to be included in the submission if reproducible download instructions are provided. All models must execute locally during inference.

6.1 Minimum Project Structure

Path	Expected contents
README.md	Provide setup and dependency installation instructions, the technical approach and key decisions, relevant thresholds and formulas, model setup details, and the exact commands required to run the solution and generate all required outputs.
src/	Source code implementing the video analytics solution and output generation
configs/	Configurable parameters, thresholds, and/or scene-specific configuration where applicable.
Dockerfile	Reproducible definition of the execution environment.
outputs/	Required CSV files and annotated video outputs.
Dependency specification	requirements.txt, pyproject.toml, or equivalent environment/dependency specification.
models/ or download instructions	Model weights if practical, otherwise reproducible download/setup instructions.

6.2 Shortlist Grading Rubric

Criterion	Weight	Assessment Focus
Task Correctness and Completeness	35%	Whether all required tasks and outputs are completed, and whether the reported results are reasonably supported by the candidate's stated methodology and applied consistently across the videos. Perfect model-level accuracy is not expected.
Technical Reasoning and Methodology	45%	Quality of the technical approach, event definitions, temporal rules, thresholds, formulas, assumptions, and justification of key decisions, as demonstrated through the documentation and written explanations.
Presentation, Written Communication, and Visualization Quality	20%	Code quality and readability, clarity and organization of the documentation, quality and readability of the output visualizations, and overall effectiveness of the written submission.
Total	100%	

6.3 Candidate Understanding and Interview Verification

Shortlisted candidates **will be required to explain and defend their implementation during the technical interview**. A candidate may be rejected if they cannot demonstrate sufficient understanding of the submitted code, methodology, or technical decisions, regardless of the take-home assessment score.